In subsequent chapters we shall return to other examples, but for now we focus on one: supersymmetry.

As we have discussed, a fundamental property of string theory is that it is highly symmetric, incorporating not only intuitive symmetry principles but respecting, as well, the maximal mathematical extension of these principles, supersymmetry. This means, as discussed in Chapter 7, that patterns of string vibrations come in pairs—superpartner pairs—differing from each other by a half unit of spin. If string theory is right, then some of the string vibrations will correspond to the known elementary particles. And due to the supersymmetric pairing, string theory makes the prediction that each such known particle will have a superpartner. We can determine the force charges that each of these superpartner particles should carry, but we do not currently have the ability to predict their masses. Even so, the prediction that superpartners exist is a generic feature of string theory; it is a property of string theory that is true, independent of those aspects of the theory we haven't yet figured out.

No superpartners of the known elementary particles have ever been observed. This might mean that they do not exist and that string theory is wrong. But many particle physicists feel that it means that the superpartners are very heavy and are thus beyond our current capacity to observe experimentally. Physicists are now constructing a mammoth accelerator in Geneva, Switzerland, called the Large Hadron Collider. Hopes run high that this machine will be powerful enough to find the superpartner particles. The accelerator should be ready for operation before 2010, and shortly thereafter supersymmetry may be confirmed experimentally. As Schwarz has said, "Supersymmetry ought to be discovered before too long. And when that happens, it's going to be dramatic."17

You should bear in mind two things, though. Even if superpartner particles are found, this fact alone will not establish that string theory is correct. As we have seen, although supersymmetry was discovered by studying string theory, it has also been successfully incorporated into point-particle theories and is therefore not unique to its stringy origins. Conversely, even if superpartner particles are not found by the Large Hadron Collider, this fact alone will not rule out string theory,